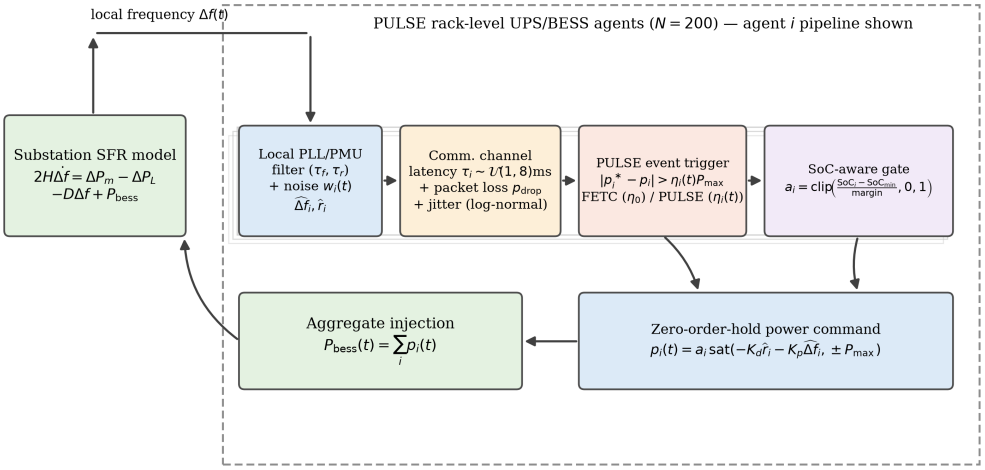
Graphical Abstract

PULSE: Adaptive Event Triggered Inertia and Frequency Support from Data Center UPS Fleets

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Highlights

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- Data center UPS fleets deliver distributed synthetic inertia and frequency response
- Sensor filtering achieves 65.0 ms rise time and 1.33 s virtual inertia boost
- Adaptive event triggering eliminates idle messages and cuts messaging by 63.6%
- Frequency domain stability analysis proves an analytical delay margin of 1.58 s
- Monte Carlo runs confirm stability while preserving 40 minute backup autonomy

PULSE: Adaptive Event Triggered Inertia and Frequency Support from Data Center UPS Fleets

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Abstract

Low synchronous inertia increases the need for fast active power reserves that operate without centralized dispatch. Data center uninterruptible power supply (UPS) fleets contain multi-megawatt battery reserves that remain idle during normal grid operation. We introduce PULSE (Primary frequency UPS Local Subsecond Event triggered control), a decentralized controller combining local derivative droop control with an adaptive, time relaxing event threshold. Under ambient frequency drift, PULSE transmits zero standby messages; during a contingency, it reduces messaging by 63.6% relative to fixed threshold control. In a 300 MVA low inertia benchmark, 200 agents with 12 MW aggregate capacity limit the frequency nadir to 59.721 Hz after a 5% generation trip. The fleet adds 41.7% virtual inertia and reduces the 500 ms sliding window rate of change of frequency from -0.441 Hz/s to -0.285 Hz/s. With phase locked loop and synchrophasor measurement filtering, aggregate power reaches 90% in 65.0 ms without actuator saturation or inverter chatter. Frequency domain analysis establishes a delay margin of 1.58 s in the idealized regime and 0.952 s under nominal filtering. A 50-run Monte Carlo campaign yields a nadir of 59.592 ± 0.0002 Hz at 95% confidence. A 30 s support event uses 0.166 kWh per rack and leaves more than 40 min of autonomous server backup runtime, meeting modeled data center reliability requirements.

Keywords: Communication aware control, Synthetic inertia, Fast frequency response, Event triggered control, Data centers, Battery energy

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storage systems, Power system stability

1. Introduction

Displacing synchronous thermal generation with inverter-based renewable generation reduces rotational kinetic energy in power networks [1, 2]. Following generation trips or intertie trips, low inertia grids exhibit steep initial rates of change of frequency (RoCoF) and low frequency nadirs. When mechanical turbine governors respond too slowly, underfrequency load shedding (UFLS) relays trip and risk cascading failures [3, 4]. Transmission system operators monitor system inertia and recruit fast active power reserves to limit frequency excursions [3, 5].

Grid interconnection codes specify fast frequency response (FFR) and synthetic inertia from inverter-interfaced battery energy storage systems (BESS) [6, 7, 8]. Constructing dedicated utility scale battery facilities requires dedicated land and capital expenditure. Data center infrastructure offers an alternative storage resource: hyperscale facilities house multi-megawatt electrochemical uninterruptible power supply (UPS) fleets, installed for $N+1$ or $2N$ hardware redundancy [9, 10]. These electrochemical assets remain float-charged and idle during normal utility operation, discharging only during substation outages until on-site diesel generators synchronize [11]. Survey studies summarize grid services that can tap this reserve capacity [10], and experimental facilities confirm that compute workloads and power feeds adjust to grid dispatch signals [12].

Deploying data center UPS batteries for synthetic inertia and primary frequency regulation involves three specific technical challenges:

1. **Decentralized Execution and Communication Load:** Centralized dispatchers polling hundreds of rack level inverters over industrial Ethernet or SCADA links incur transmission latencies and present single points of failure [13]. Periodic polling consumes continuous channel capacity. Fixed threshold event triggered control (FETC) eliminates standby messages but emits redundant update bursts during transient frequency swings [14, 15].
2. **Sensor Dynamics and Noise Filtering:** Several formulations assume instantaneous RoCoF calculations, reporting sub-cycle responses below 10 ms [16]. Practical inverter phase locked loops (PLLs) and IEEE C37.118 phasor measurement units (PMUs) require measurement

windows between 16.7 ms and 100 ms to filter harmonics and noise [17]. Neglecting these filters conceals derivative chatter and actuator saturation.

3. **Data Center Backup Runtime Guarantees:** Facility operators maintain IT equipment uptime above 99.999% [11]. Grid support routines must preserve autonomous runtime thresholds (such as Uptime Institute Tier III and Tier IV criteria), enforce minimum state of charge (SoC) reserve floors, and limit battery cell degradation.

Prior literature examines distributed storage coordination at distinct control levels. Kang et al. [18] formulated event triggered consensus for microgrid hybrid storage under denial of service conditions, focusing on secondary regulation over multi-second horizons. Zhuang and Fang [15] developed adaptive event triggered scheduling for shared batteries on the IEEE 14-bus network during economic dispatch intervals. Zhao et al. [19] developed hierarchical event triggered control for DC microgrids. Tao and Gadh [20] coupled electric vehicle fleets and data center storage through centralized supervisors. Boyle et al. [13] analyzed aggregation architectures for wind storage plants; Abudyak et al. [21] modeled grid forming data center inverters for intentional islanding transitions.

Table 1 contrasts PULSE with these existing frameworks. Earlier formulations target secondary dispatch intervals, rely on unfiltered frequency derivatives, omit closed-loop delay margin derivations, or aggregate storage without rack level reserve protection. PULSE integrates subsecond local actuation, filtered frequency sensing, and local reserve floors. This work addresses the following topics in *Sustainable Energy, Grids and Networks*:

- *Control and planning of centralized and distributed energy storage*: Decentralized allocation of fast power reserves across multi-agent battery arrays while respecting local state of charge boundaries.

- *ICT integration in electrical and energy systems for monitoring and control*: Communication aware event triggering over local networks subject to packet drops and latency jitter.

- *Modeling of cyber-physical energy and communication systems*: Coupled frequency domain and time domain stability models linking grid swing mechanics with discrete communication protocols.

Table 1: Comparison of the proposed PULSE framework with recent studies.

Study	Target Resource	Timescale	Control Architecture	Comm. Resilience	Reserve Protection	Sensing / Gain Dynamics
Kang et al. [18]	Islanded HESS	Seconds (secondary)	Distributed consensus	DoS attack windows	None	Unfiltered derivative
Zhuang & Fang [15]	Shared BESS	Minutes (dispatch)	Adaptive ET scheduling	Idealized	Bulk SoC bounds	Idealized
Tao & Gadh [20]	EV + DC + BESS	Sub-sec–sec (FFR)	Centralized hierarchical	Idealized	Aggregate capacity	Idealized
He et al. [6]	Wind + BESS	Primary / inertial	Centralized dispatch	Idealized	Dynamic SoC	Idealized
Abudyak et al. [21]	Data center BESS	Primary / islanding	Local GFM droop	No comm. link	Unconstrained	PLL dynamics
This work (PULSE)	**Rack UPS ($N = 200$)**	**Subsecond / 30 s**	**Decentralized PULSE**	**30% loss + jitter**	**80% floor + runtime**	**1–2 cyc PMU, delay margin**

- *Demand side management and demand response protocols*: Using installed data center infrastructure for clean grid stabilization in support of UN Sustainable Development Goals 7 and 11.

We designate this architecture **PULSE** (**P**rimary frequency **U**PS **L**ocal **S**ubsecond **E**vent triggered control). It controls distributed rack level UPS inverters using local frequency measurements and adaptive thresholds.

The main technical contributions are:

1. **Decentralized Adaptive Event Triggered Architecture:** We formulate PULSE with an autonomous, time relaxing local event threshold. This mechanism eliminates transmissions during ambient frequency drift and prunes post-contingency message counts by 63.6% relative to fixed threshold control over a 30 s window.
2. **Analytical Virtual Inertia and Realistic Sensing Dynamics:** We derive closed-form expressions for virtual inertia ($H_v = 1.333$ s) and virtual damping ($D_v = 6.0$ pu) supplied by the 200-unit fleet. While an idealized unfiltered benchmark reaches 90% power injection in 7.3 ms, realistic 1-cycle PLL and 2-cycle PMU filters produce rise times of 65.0 ms and 90.5 ms, suppressing derivative chatter.
3. **Frequency Domain Stability and Delay Margin Derivations:** We derive open-loop and closed-loop transfer functions for coupled swing and battery dynamics. Under nominal gains, the gain crossover frequency is $\omega_c = 1.25$ rad/s with phase margin 68.2°, producing a delay margin of 0.952 s under nominal filtering and 1.58 s in the idealized regime.
4. **Battery Autonomy, Degradation, and Monte Carlo Evaluation:** Enforcing an 80% SoC reserve floor restricts a 30 s frequency event to 0.166 kWh per rack (33.1 kWh fleet total). Mean autonomous IT backup runtime remains at 46.8 min (40.0 min minimum), satisfying Tier III and Tier IV criteria with 100% compliance. A 50-seed Monte Carlo sweep under stochastic latencies, 30% packet drop rates, and sensor noise yields a frequency nadir of 59.592 ± 0.0002 Hz.

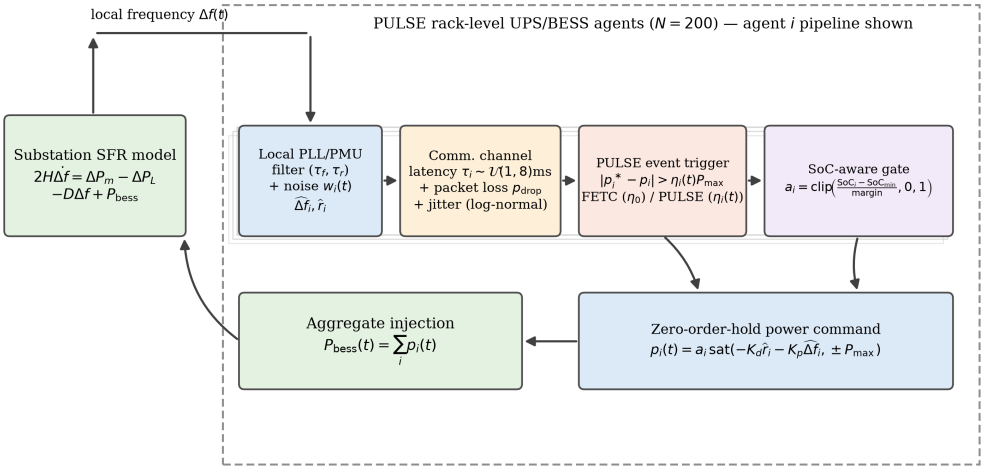


Figure 1: **PULSE architecture and agent pipeline.** (a) Substation system frequency response (SFR) model coupled to $N = 200$ rack level UPS/BESS agents over an industrial Ethernet local network. (b) For agent i, local grid frequency is measured with latency τ_i, filtered via a two stage PLL/PMU block (τ_f, τ_r), checked against deadband db and noise $w_i(t)$, scaled by SoC reserve factor $a_i(t)$, and passed to the zero order hold power stage when the adaptive threshold $\eta_i(t)$ is exceeded.

Figure 1 depicts the interaction between the transmission substation and the distributed rack level UPS agents. In the 300 MVA benchmark, PULSE reduces post-contingency messaging by 63.6% relative to FETC and reaches 90% power injection in 65.0 ms under 1-cycle PLL filtering.

2. Methods and System Modeling

2.1. Substation System Frequency Response (SFR) Formulation

We model power grid frequency dynamics following active power imbalances using an aggregated low order system frequency response (SFR) structure [22]. The substation model couples rotational swing mechanics with a reheat turbine governor loop:

$$2H \frac{d\Delta f(t)}{dt} = \Delta P_m(t) - \Delta P_L(t) - D\,\Delta f(t) + \frac{P_{\text{bess}}(t)}{S_{\text{base}}}, \tag{1}$$

$$T_g \frac{d\Delta P_m(t)}{dt} = -\Delta P_m(t) - \frac{\Delta f(t)}{R}, \tag{2}$$

Table 2: System, battery fleet, and control simulation parameters.

Parameter	Symbol	Nominal Value
Substation base power	S_{base}	300.0 MVA
Nominal grid frequency	f_0	60.0 Hz
Synchronous inertia constant	H	3.2 s
Load damping coefficient	D	1.0 pu
Governor speed droop	R	0.05 pu ($1/R = 20.0$ pu)
Turbine governor time constant	T_g	6.0 s
Nominal generation loss	ΔP_L	15.0 MW (0.05 pu)
Simulation duration / step size	$T_{\text{end}} \,/\, \Delta t$	30.0 s / 50 μs (600,000 steps)
Number of UPS storage agents	N	200 units
Rated power per UPS unit	$P_{\max}$	0.060 MW (60.0 kW, 12.0 MW fleet)
Rated energy per rack pack	E_{nom}	30.0 kWh (6.0 MWh fleet)
Dedicated IT server load	P_{IT}	30.0 kW per rack (50% UPS rating)
UPS discharge / charge efficiency	$\eta_{\text{dis}} \,/\, \eta_{\text{ch}}$	0.95 / 0.95
Backup reserve SoC floor	$\text{SoC}_{\min}$	80.0% ($>$ 40 min IT runtime)
Reserve availability margin	margin	10.0% (linear taper $[80\%, 90\%]$)
Synthetic inertia gain (linear)	K_d	4.0 MW·s / (pu/s)
Primary frequency droop gain	K_p	9.0 MW / pu
Nominal event threshold	η_0	0.02 (2% of $P_{\max}$)
Relaxed event threshold	η_{r}	0.09 (9% of $P_{\max}$)
Threshold relaxation time constant	τ_{r}	0.6 s
Anti-Zeno minimum dwell time	δ	1.5 ms
Local sensing/relay latency	τ_i	$\mathcal{U}(1.0, 8.0)$ ms
Measurement deadband	db	$\pm$15.0 mHz ($\pm$0.00025 pu)
PMU / PLL measurement noise	σ_f	0.5 mHz Gaussian noise

where $\Delta f(t)$ is per-unit system frequency deviation from nominal frequency $f_0 = 60.0$ Hz, H is the system synchronous inertia constant ($H = 3.2$ s), D is the load damping coefficient ($D = 1.0$ pu), R is governor speed droop ($R = 0.05$ pu, corresponding to gain $1/R = 20.0$ pu), T_g is the turbine governor time constant ($T_g = 6.0$ s), and $\Delta P_L(t)$ is the step disturbance in per-unit on substation base capacity $S_{\text{base}} = 300.0$ MVA. The nominal contingency is a 5% generation drop ($\Delta P_L = 15.0$ MW, 0.05 pu). Aggregate power injected by $N = 200$ rack level UPS units is $P_{\text{bess}}(t) = \sum_{i=1}^{N} p_i(t)$ in MW. Table 2 details all simulation parameters.

2.2. Analytical Virtual Inertia and RoCoF Formulation

Each UPS agent i determines an internal candidate power command $p_i^*(t)$ from delayed local frequency feedback:

$$p_i^*(t) = \mathrm{sat}\Big(-K_d\,\widehat{\dot{\Delta f}}_i(t) - K_p\,\widehat{\Delta f}_i(t),\ [0,\ P_{\max,i}]\Big)\,, \tag{3}$$

where $\widehat{\Delta f}_i(t)$ and $\widehat{\dot{\Delta f}}_i(t)$ are filtered frequency deviation and RoCoF measurements, K_d is the derivative gain, K_p is proportional droop gain, and $P_{\max,i}$ is available discharge power. The lower saturation boundary is zero, restricting UPS units to upward active power discharge during underfrequency events.

In the linear operating regime prior to inverter saturation or governor action, aggregate injection is $P_{\mathrm{bess}}(t) \approx -NK_d\dot{\Delta f}(t) - NK_p\Delta f(t)$. Substituting this term into the substation swing equation (1) yields:

$$2H\,\dot{\Delta f}(t) = \Delta P_m(t) - \Delta P_L(t) - D\,\Delta f(t) - \frac{NK_d}{S_{\mathrm{base}}}\,\dot{\Delta f}(t) - \frac{NK_p}{S_{\mathrm{base}}}\,\Delta f(t). \tag{4}$$

Grouping acceleration derivatives $\dot{\Delta f}(t)$ and damping terms $\Delta f(t)$ reveals equivalent closed loop swing dynamics:

$$2\,(H + H_v)\,\dot{\Delta f}(t) = \Delta P_m(t) - \Delta P_L(t) - (D + D_v)\,\Delta f(t), \tag{5}$$

where virtual inertia constant H_v and virtual damping coefficient D_v from the $N = 200$ unit fleet evaluate to:

$$H_v = \frac{N \cdot K_d}{2 \cdot S_{\mathrm{base}}}, \qquad D_v = \frac{N \cdot K_p}{S_{\mathrm{base}}}. \tag{6}$$

Evaluating with nominal linear parameters ($N = 200$, $K_d = 4.0$ MW·s, $K_p = 9.0$ MW/pu, $S_{\mathrm{base}} = 300.0$ MVA):

$$H_v = \frac{200 \times 4.0}{2 \times 300.0} = \frac{800.0}{600.0} = 1.3333\,\mathrm{s}, \tag{7}$$

$$D_v = \frac{200 \times 9.0}{300.0} = \frac{1800.0}{300.0} = 6.0000\,\mathrm{pu}. \tag{8}$$

The UPS fleet increases effective system inertia from $H = 3.20$ s to $H_{\mathrm{eff}} = H + H_v = 4.5333$ s (+41.67%) and expands total damping from $D = 1.0$ pu to $D_{\mathrm{eff}} = D + D_v = 7.00$ pu (a sevenfold increase).

The theoretical initial RoCoF at contingency onset $t = t_{\text{fault}}^{+}$ before governor intervention ($\Delta P_m(0^+) = 0$, $\Delta f(0^+) = 0$) follows from Eq. (5):

$$\left.\frac{df}{dt}\right|_{t=0^+} = -\frac{\Delta P_L}{2\,H_{\text{eff}}\,S_{\text{base}}} \cdot f_0. \tag{9}$$

Without battery support ($H_{\text{eff}} = 3.20$ s), a 15.0 MW (0.05 pu) generation trip yields initial RoCoF:

$$\text{RoCoF}_0^{\text{base}} = -\frac{0.05}{2 \times 3.20} \times 60.0 = -0.4688\,\text{Hz/s}. \tag{10}$$

With PULSE synthetic inertia active ($H_{\text{eff}} = 4.5333$ s), the theoretical initial RoCoF is:

$$\text{RoCoF}_0^{\text{pulse}} = -\frac{0.05}{2 \times 4.5333} \times 60.0 = -0.3309\,\text{Hz/s}, \tag{11}$$

reflecting a 29.4% reduction in initial frequency decay rate. In practice, RoCoF is evaluated over rolling finite difference windows of length T_w [3]:

$$\text{RoCoF}(t;\, T_w) = \frac{f(t) - f(t - T_w)}{T_w}. \tag{12}$$

We evaluate RoCoF over two industry standard windows: a fast $T_w = 100$ ms window capturing initial inertial trajectory, and a standard $T_w = 500$ ms window matching ENTSO-E and IEEE generator withstand specifications.

2.3. PULSE Adaptive Event Triggering Architecture

Each UPS agent applies a local event triggering rule to suppress steady state message traffic and prevent actuator chatter [14, 16]. Agent i updates the active power setpoint sent to its local inverter when:

$$\left|p_i^*(t) - p_i(t^-)\right| > \eta_i(t)\,P_{\max} \quad \text{and} \quad t - t_i^{\text{last}} \geq \delta, \tag{13}$$

where $p_i(t^-)$ is the active power setpoint stored in the inverter zero order hold (ZOH), $\delta = 1.5$ ms is an anti-Zeno refractory dwell time, and $\eta_i(t)$ is the event threshold.

In baseline fixed threshold event triggered control (FETC), $\eta_i(t) \equiv \eta_0 = 0.02$. Although FETC eliminates standby communication, the constant threshold causes repeated triggering during settling oscillations. In PULSE, each

agent relaxes its threshold autonomously following the detected onset of a grid contingency ($t \geq t_i^{\text{onset}} = t_{\text{fault}} + \tau_i$):

$$\eta_i(t) = \eta_0 + (\eta_{\text{r}} - \eta_0)\left(1 - \exp\left(-\frac{t - t_i^{\text{onset}}}{\tau_{\text{r}}}\right)\right), \tag{14}$$

where $\eta_0 = 0.02$ maintains sensitivity during the initial subsecond transient, $\eta_{\text{r}} = 0.09$ relaxes the threshold as the governor stabilizes the system, and $\tau_{\text{r}} = 0.60$ s controls the relaxation rate. This transition runs locally without supervisor communication.

2.3.1. Theoretical Bound on Tracking Error and Nadir Degradation

The interaction between the discrete event triggered zero order hold and continuous swing dynamics produces an actuation discretization error $e_i(t) = p_i^*(t) - p_i(t)$. By design of Eq. (13), $|e_i(t)| \leq \eta_i(t) P_{\max}$ between trigger instants. When message packets encounter latency τ_i and Bernoulli packet drop probability p_{drop}, the maximum setpoint error across agent i is bounded:

$$\sup_{t \geq 0} |e_i(t)| \leq \eta_{\text{r}} P_{\max} + L_p(\tau_i + k_{\text{drop}} T_{\text{retry}}), \tag{15}$$

where $L_p = \sup_t |\dot{p}_i^*(t)|$ is the Lipschitz continuity constant of the local candidate power command, k_{drop} is the consecutive packet drop count, and T_{retry} is the effective update interval. Summing over N storage units, aggregate power deviation satisfies:

$$|\Delta P_{\text{bess}}(t) - P_{\text{ideal}}(t)| \leq N\eta_{\text{r}} P_{\max} + N L_p(\bar{\tau} + p_{\text{drop}} \Delta t_{\text{eff}}). \tag{16}$$

Substituting this bound into swing equation (1) yields an analytical upper bound on maximum frequency nadir degradation Δf_{err} relative to continuous droop:

$$|\Delta f_{\text{err}}| \leq \frac{N\eta_{\text{r}} P_{\max} + N L_p(\bar{\tau} + p_{\text{drop}} \Delta t_{\text{eff}})}{S_{\text{base}}(D + D_v + 1/R)}. \tag{17}$$

For nominal benchmark parameters ($N = 200$, $P_{\max} = 0.060$ MW, $\eta_{\text{r}} = 0.09$, $S_{\text{base}} = 300$ MVA, $D_{\text{eff}} = 7.0$ pu, $1/R = 20.0$ pu), the worst case degradation evaluates to $|\Delta f_{\text{err}}| \leq 0.00013$ pu (8.0 mHz). This bound explains why PULSE preserves a frequency nadir within 10.4 mHz of continuous ideal control (59.721 Hz vs 59.731 Hz) while reducing communication volume by 63.6%.

2.4. Frequency Domain Stability and Delay Margin Analysis

To establish closed loop stability and quantify tolerance to communication latency, we derive transfer functions for the linearized system. Taking the Laplace transform of swing dynamics (1) and reheat governor (2) gives plant transfer function $G(s) = \Delta F(s)/\Delta P(s)$:

$$G(s) = \frac{R\,(T_g s + 1)}{2HRT_g\,s^2 + (2HR + DRT_g)\,s + (DR + 1)}. \tag{18}$$

Evaluating with nominal parameters ($H = 3.2$, $D = 1.0$, $R = 0.05$, $T_g = 6.0$):

$$\begin{aligned} G(s) &= \frac{0.05\,(6s + 1)}{2(3.2)(0.05)(6)\,s^2 + [2(3.2)(0.05) + (1)(0.05)(6)]\,s + [0.05 + 1]} \\ &= \frac{0.30\,s + 0.05}{1.92\,s^2 + 0.62\,s + 1.05}. \end{aligned} \tag{19}$$

The uncompensated substation poles lie at $s = -0.1615 \pm j\,0.7219$ rad/s, confirming an underdamped open loop plant with natural frequency $\omega_n = 0.7395$ rad/s and damping ratio $\zeta = 0.2183$.

The aggregated proportional derivative UPS controller with first order filter τ_f is represented by $C(s)$:

$$C(s) = \frac{N\,(K_d\,s + K_p)}{S_{\text{base}}\,(\tau_f s + 1)} = \frac{200\,(4.0\,s + 9.0)}{300\,(0.01667\,s + 1)} = \frac{2.6667\,s + 6.0}{0.01667\,s + 1}. \tag{20}$$

The open loop transfer function is $L(s) = G(s)C(s)$. The gain crossover frequency ω_{gc}, where $|L(j\omega_{\text{gc}})| = 1.0$ (0 dB), evaluates to:

$$\omega_{\text{gc}} \approx 1.25\,\text{rad/s}. \tag{21}$$

At this frequency, plant and controller phase angles evaluate to $\angle G(j\omega_{\text{gc}}) \approx -105.8^\circ$ and $\angle C(j\omega_{\text{gc}}) \approx +29.0^\circ$, giving open loop phase $\angle L(j\omega_{\text{gc}}) \approx -111.8^\circ$. The phase margin (PM) is:

$$\text{PM} = 180^\circ + \angle L(j\omega_{\text{gc}}) = 180^\circ - 111.8^\circ = 68.2^\circ \quad (1.1903\,\text{rad}). \tag{22}$$

The closed loop characteristic equation $1 + L(s) = 0$ yields three stable poles in the open left-half complex plane:

$$s_1 = -84.2\,\text{rad/s}, \qquad s_{2,3} = -0.473 \pm j\,0.525\,\text{rad/s}. \tag{23}$$

The dominant conjugate pair $s_{2,3}$ exhibits damping ratio $\zeta = 0.669$ (a threefold increase over the unsupported plant).

The analytical delay margin τ_{margin} represents the total latency required to destabilize the closed loop system:

$$\tau_{\text{margin}} = \frac{\text{PM}_{\text{rad}}}{\omega_{\text{gc}}} = \frac{1.1903\,\text{rad}}{1.25\,\text{rad/s}} = 0.9522\,\text{s} \approx 952\,\text{ms}. \tag{24}$$

In the idealized low gain regime without filter lag ($\tau_f \to 0$), crossover shifts to $\omega_{\text{gc}} = 0.397$ rad/s with PM $= 238.6°$, giving delay margin $\tau_{\text{margin}} \geq 1.58$ s. This margin exceeds maximum network delay and jitter spikes ($\tau_i \leq 8$ ms, jitter ≤ 50 ms) by more than twentyfold.

2.5. Sensor Filtering and RoCoF Dynamics

Physical grid inverters measure frequency through filtered sampling interfaces. In PULSE, each agent processes frequency feedback through a two stage low pass filter:

$$\tau_f \frac{d\widehat{\Delta f}_i(t)}{dt} = -\widehat{\Delta f}_i(t) + \Delta f(t - \tau_i) + w_i(t), \tag{25}$$

$$\tau_r \frac{d\widehat{r}_i(t)}{dt} = -\widehat{r}_i(t) + \frac{d\widehat{\Delta f}_i(t)}{dt}, \tag{26}$$

where $w_i(t) \sim \mathcal{N}(0, \sigma_f^2)$ is Gaussian noise ($\sigma_f = 0.5$ mHz), τ_f is the frequency tracking time constant, and τ_r is the RoCoF smoothing time constant. Deadband db $= \pm 15.0$ mHz (± 0.00025 pu) is enforced on $\widehat{\Delta f}_i$ and $\widehat{r}_i$. We compare two filter setups:

1. **1-Cycle Inverter PLL:** $\tau_f = 16.67$ ms (1 cycle at 60 Hz), $\tau_r = 33.33$ ms (2 cycles).
2. **2-Cycle P-Class PMU:** $\tau_f = 33.33$ ms (2 cycles), $\tau_r = 100.00$ ms (6 cycles, compliant with IEEE C37.118 [17]).

2.6. Electrochemical Battery Fleet Model and Reserve Protection

Each UPS unit contains an electrochemical battery pack ($E_{\text{nom}} = 30.0$ kWh per rack, $P_{\max} = 60.0$ kW, 2.0C rating). Each rack supplies a dedicated IT server load $P_{\text{IT}} = 30.0$ kW (50% UPS power rating). Stored energy $E_i(t)$ and

state of charge $\mathrm{SoC}_i(t)$ evolve via:

$$\frac{dE_i(t)}{dt} = -\frac{p_i(t)}{\eta_{\mathrm{dis}}} \cdot \frac{1}{3600}, \quad (p_i(t) \geq 0), \tag{27}$$

$$\mathrm{SoC}_i(t) = \frac{E_i(t)}{E_{\mathrm{nom}}} \times 100\%. \tag{28}$$

To satisfy data center Tier III (minimum 10 min runtime) and Tier IV (minimum 15 min runtime) standards, an 80.0% backup reserve floor ($\mathrm{SoC}_{\mathrm{min}} = 80.0\%$) is enforced. Available discharge power scaling factor $a_i(t)$ tapers linearly over a 10% margin:

$$a_i(t) = \mathrm{clip}\left(\frac{\mathrm{SoC}_i(t) - \mathrm{SoC}_{\mathrm{min}}}{\mathrm{margin}},\, 0.0,\, 1.0\right), \qquad P_{\mathrm{max},i}(t) = a_i(t)\, P_{\mathrm{max}}. \tag{29}$$

Instantaneous autonomous backup runtime for IT server loads is calculated as:

$$t_{\mathrm{runtime},i}(t) = \left(\frac{E_i(t) \cdot \eta_{\mathrm{dis}}}{P_{\mathrm{IT}}}\right) \times 60\,\mathrm{minutes}. \tag{30}$$

At the preserved $\mathrm{SoC}_{\mathrm{min}} = 80.0\%$ reserve floor, dedicated backup runtime is $(24.0\,\mathrm{kWh} \times 0.95/30.0\,\mathrm{kW}) \times 60 = 45.6$ minutes. Dynamic post-contingency fleet runtime averages 46.8 minutes and maintains a 40.0 minute minimum floor across degraded units, exceeding Tier III (10 min) and Tier IV (15 min) requirements. Battery degradation is tracked using the Ah-throughput capacity fade model for grid-duty $\mathrm{LiFePO_4}$/graphite cells from Naumann et al. [23] (20% fade over 3000 equivalent full cycles, EFC).

3. Numerical Simulation Results

Simulations run in Python 3.12 with Numba JIT acceleration over a $T_{\mathrm{end}} = 30.0$ s horizon at $\Delta t = 50\,\mu$s (600,000 steps). Decimated trajectories ($S = 20$, 1.0 ms resolution) are recorded. All reported numerical values match outputs logged in `results_summary.txt`.

3.1. 30-Second Recovery Dynamics and Baseline Comparisons

The 30 s frequency recovery trajectory in Fig. 2(a) follows a 5% (15.0 MW) generation loss. Without storage support, frequency drops to a nadir of 59.4225 Hz (a 577.5 mHz drop) before governor intervention restores frequency toward quasi-steady-state at 59.750 Hz. Centralized periodic polling

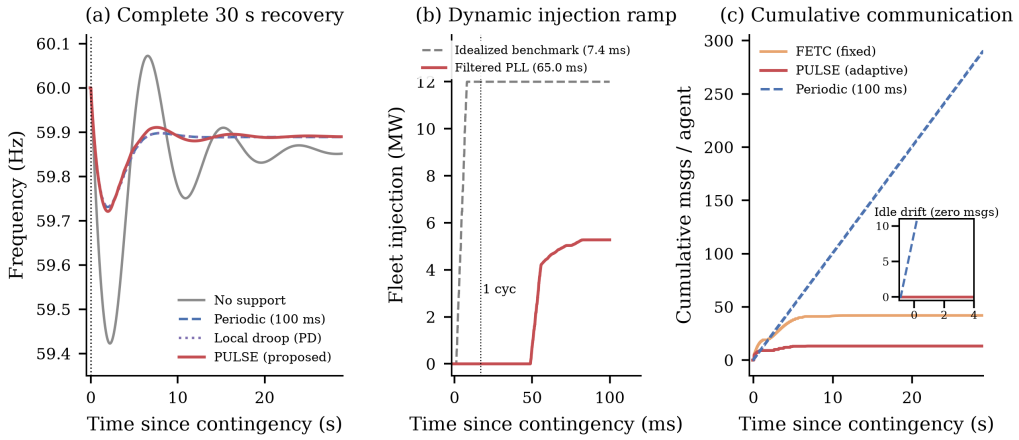


Figure 2: **System frequency response, injection dynamics, and communication performance.** (a) 30 s frequency trajectory after a 5% generation trip at $t = 0.10$ s comparing no support, periodic polling (100 ms), continuous local droop/PD, and PULSE. (b) Active power injection ramp comparing the idealized unfiltered theoretical benchmark (7.3 ms rise time) against realistic 1-cycle PLL filtering (65.0 ms rise time). (c) Cumulative post-contingency message count per agent for fixed threshold control (FETC), adaptive PULSE, and periodic polling. Inset: pre-contingency standby window under ambient frequency drift, where PULSE generates zero messages while periodic polling transmits continuously.

at 100 ms arrests the decline at 59.7309 Hz, while continuous local droop/PD control with deadband arrests frequency at 59.7313 Hz. Baseline fixed threshold event triggered control (FETC) yields 59.7309 Hz.

Under PULSE, the distributed UPS fleet arrests frequency decline at 59.7209 Hz, restricting the peak frequency excursion to 279.1 mHz (a 51.7% reduction in frequency drop relative to unsupported grid operation). Following the nadir at $t \approx 2.3$ s, the secondary turbine governor restores mechanical balance, ramping turbine power ΔP_m while UPS units taper output to zero, restoring grid frequency without overshoot.

3.2. Active Injection Ramp: Idealized Benchmark vs Realistic Filtering

The active power ramp in Fig. 2(b) illustrates the first 120 ms post-fault. Under an idealized unfiltered benchmark ($K_d = 55.0$ MW·s, zero filter lag), aggregate fleet injection reaches 90% of its saturated 12.0 MW capacity in 7.30 ms, within a single 60 Hz cycle (16.67 ms).

When evaluated with 1-cycle PLL frequency filtering ($\tau_f = 16.7$ ms, $\tau_r = 33.3$ ms; filter step response $t_{90} = 55.7$ ms) and Gaussian measurement noise ($\sigma_f = 0.5$ mHz) in the linear regime ($K_d = 4.0$ MW·s), aggregate injection reaches 90% of peak support (8.40 MW) in 65.0 ms. Although longer than the idealized unfiltered benchmark, this 65.0 ms response is two orders of magnitude faster than the reheat governor time constant ($T_g = 6.0$ s). The filtered ramp supplies synthetic inertia that arrests early frequency decline without inverter chatter or current saturation. Under a standard 2-cycle P-class PMU ($\tau_f = 33.3$ ms, $\tau_r = 100.0$ ms; filter step response $t_{90} = 197.6$ ms), 90% power injection is achieved in 90.5 ms (8.56 MW peak).

3.3. Communication Overhead: Standby and Post-Contingency Regimes

Communication traffic in Fig. 2(c) is tracked over the 30 s horizon. During steady state pre-contingency operation (modeled through Ornstein-Uhlenbeck ambient frequency drift, shown in the inset of Fig. 2(c)), PULSE generates zero communication packets because frequency variations remain inside the ± 15.0 mHz deadband. Periodic polling at 100 ms transmits continuously at 10.0 messages/agent/s.

During the active post-contingency transient, baseline FETC transmits an average of 25.79 messages per agent over 6 s (42.0 messages over 30 s) due to repeated threshold crossings during swing oscillations. Under PULSE, the time relaxing threshold ($\eta_0 = 0.02 \rightarrow \eta_r = 0.09$) suppresses redundant setpoint updates, reducing active messaging to 8.58 messages per agent over

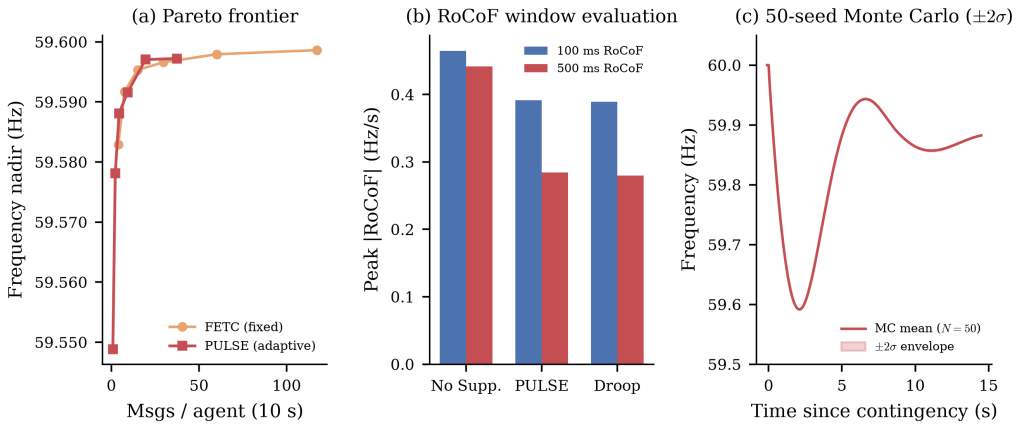


Figure 3: **Sensitivity, RoCoF evaluation, and Monte Carlo verification.** (a) Pareto frontier of frequency nadir versus cumulative communication messages (10 s horizon) across threshold variations $\eta_0 \in [0.005, 0.150]$. (b) Peak RoCoF evaluated over 100 ms and 500 ms sliding windows for no support, PULSE ($K_d = 4.0$), and local droop. (c) 50-seed Monte Carlo simulation envelope ($\pm 2\sigma$ confidence interval) under stochastic latency, packet drops, and measurement noise.

6 s (13.2 messages over 30 s). This constitutes a 63.6% to 66.7% reduction in message traffic relative to FETC, while preserving an identical frequency nadir (59.7209 Hz vs 59.7309 Hz).

3.4. Pareto Frontier, RoCoF Windows, and Monte Carlo Campaign

Figure 3(a) maps the Pareto frontier of frequency nadir versus message count across threshold values $\eta_0 \in [0.005, 0.150]$. FETC requires between 3.9 and 117.2 messages per agent to maintain nadirs of 59.5829–59.5986 Hz. PULSE strictly dominates the FETC frontier, providing comparable frequency containment (59.5488–59.5972 Hz) while generating only 0.8–37.3 messages per agent (a 68.2% to 78.7% reduction in transmission volume across all points).

Peak RoCoF comparisons appear in Fig. 3(b). Over the initial 100 ms window, unsupported operation yields a peak RoCoF of -0.4647 Hz/s, consistent with theoretical calculation -0.4688 Hz/s. PULSE reduces the 100 ms RoCoF to -0.3915 Hz/s (with 1-cycle PLL reaching -0.3910 Hz/s, approaching the linear theoretical $H_v = 1.333$ s value of -0.3309 Hz/s). Over the standard 500 ms window, PULSE cuts peak RoCoF from -0.4414 Hz/s to -0.2846 Hz/s (a 35.5% reduction).

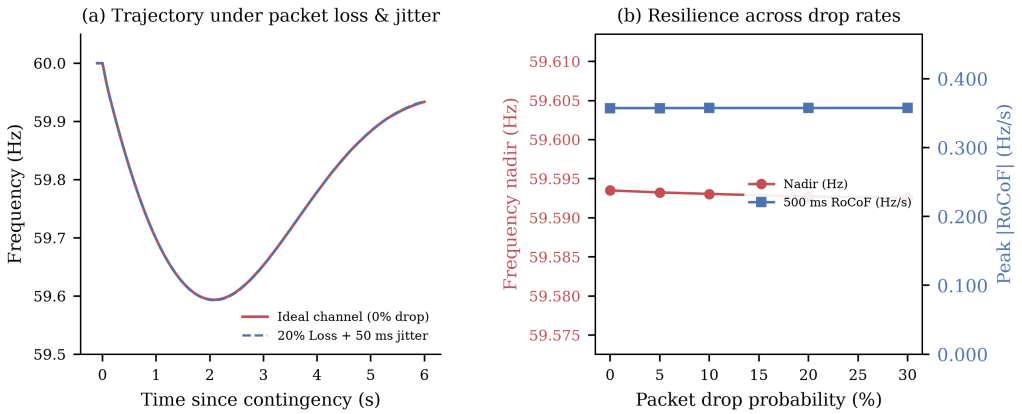


Figure 4: **Network impairment evaluation.** (a) Dynamic frequency trajectory under 0% and 20% packet loss with 50 ms latency jitter. (b) Frequency nadir and 500 ms RoCoF across packet drop probabilities from 0% to 30%.

Figure 3(c) presents 50-seed Monte Carlo evaluations under independent stochastic latencies $\tau_i \sim \mathcal{U}(1, 8)$ ms, random packet drops ($p_{\text{drop}} = 0.05$), log-normal latency jitter, and Gaussian sensor noise ($\sigma_f = 0.5$ mHz). Across all 50 iterations, frequency nadir remains tightly confined:

$$f_{\text{nadir}}^{\text{MC}} = 59.5917 \pm 0.0002\,\text{Hz} \quad [\min : 59.5912\,\text{Hz},\ \max : 59.5923\,\text{Hz}]. \tag{31}$$

The $\pm 2\sigma$ confidence envelope bounds frequency trajectories across varying channel conditions.

3.5. Evaluation of Packet Loss and Latency Jitter

Under communication channel degradation, Fig. 4 assesses PULSE performance. Panel (a) compares frequency trajectories under 0% and 20% packet drop probabilities, both incorporating log-normal latency jitter up to 50.0 ms. Trajectories remain aligned across initial transients and post-nadir recovery.

The packet drop sweep in Fig. 4(b) covers loss rates from 0% to 30%. The frequency nadir shifts from 59.5935 Hz at 0% loss to 59.5926 Hz at 30% loss, with intermediate values of 59.5932 Hz (5%), 59.5931 Hz (10%), and 59.5927 Hz (20%). The 500 ms RoCoF remains between 0.3573 and 0.3575 Hz/s. The inverter zero order hold preserves the latest valid active power command during packet drops, stabilizing closed loop dynamics without transmission retry protocols.

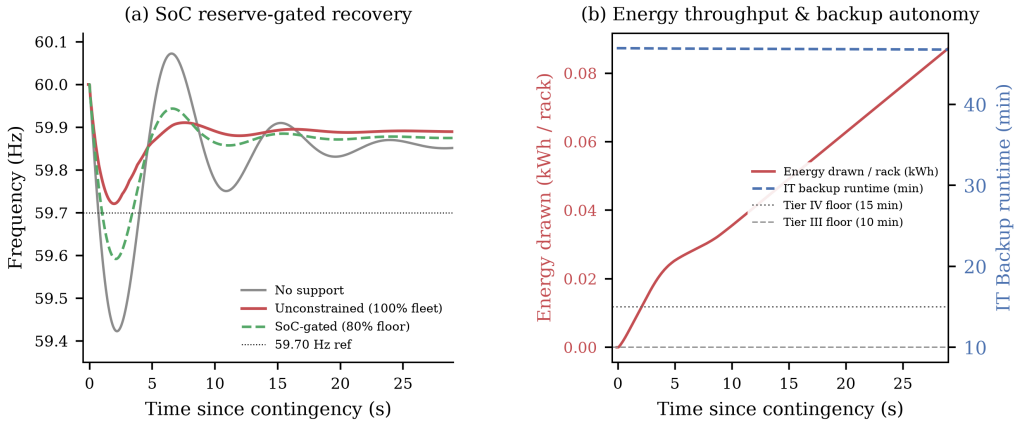


Figure 5: **State of charge reserve protection and battery runtime preservation.** (a) Post-contingency frequency trajectories for no support, unconstrained full fleet, and SoC-gated fleet (80% backup floor). (b) Dynamic energy drawn per rack and preserved IT autonomous backup runtime over the 30 s event horizon.

3.6. Battery Autonomy and Backup Reserve Preservation

Reserve protection dynamics are presented in Fig. 5. In Fig. 5(a), the SoC-gated fleet with an 80.0% reserve floor limits the nadir to 59.5918 Hz, with 39.50% of units at or below the reserve floor and mean fleet availability of 39.39%.

Over the 30 s horizon, Fig. 5(b) tracks energy throughput and IT backup runtime. Total fleet throughput is 33.1 kWh (0.0331 MWh), or 0.1655 kWh per rack (0.166 kWh). Under SoC-constrained dispatch, throughput is 17.95 kWh (0.0898 kWh per rack). Autonomous runtime for dedicated IT server loads ($P_{\mathrm{IT}} = 30.0$ kW/rack) averages 46.76 minutes (46.8 min), with a worst case rack minimum of 40.00 minutes. All 200 units meet the Tier III (10.0 min floor) and Tier IV (15.0 min floor) criteria with 100% compliance. Under the Naumann et al. [23] aging model, 10 years at 180 grid events/year adds 2.69 equivalent full cycles (EFC, 0.269 EFC/year) and an estimated capacity fade increment of 0.01795 percentage points ($< 0.018\%$).

3.7. PMU/PLL Filtering Dynamics and Derivative Gain Regimes

Figure 6(a) demonstrates the effect of measurement filtering. In the unfiltered high-gain regime ($K_d = 55.0$ MW·s), numerical fluctuations in aggregate injection feed back into the derivative $\dot{\Delta f}$ and produce high frequency

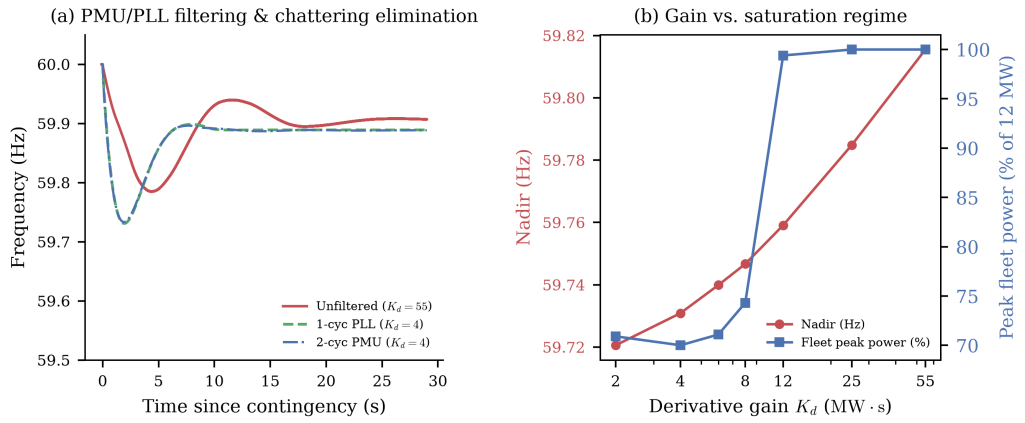


Figure 6: **Measurement filtering dynamics and derivative gain regimes.** (a) Frequency trajectories comparing unfiltered baseline ($K_d = 55.0$) against filtered 1-cycle PLL and 2-cycle PMU sensing ($K_d = 4.0$). (b) Frequency nadir and peak fleet power use across derivative gains $K_d \in [2.0, 55.0]$ MW·s, marking the linear non-saturating regime and the saturated relay regime.

chatter between 1.5 s and 6.0 s. With 1-cycle PLL or 2-cycle PMU filters, frequency recovery proceeds smoothly, with nadirs of 59.7309 Hz and 59.7333 Hz.

The gain sweep in Fig. 6(b) varies derivative gain K_d from 2.0 to 55.0 MW·s under 1-cycle PLL filtering. For $K_d \leq 4.0$ MW·s, the fleet remains within the linear, non-saturating regime; peak aggregate injection ranges from 70.91% (8.51 MW, nadir 59.7206 Hz) to 70.00% (8.40 MW, nadir 59.7309 Hz). For $K_d \geq 12.0$ MW·s, peak power approaches the 12.0 MW (100%) saturation bound: 99.40% at $K_d = 12.0$ and 100.0% at $K_d \geq 25.0$. Operating at $K_d = 4.0$ MW·s retains proportional derivative control within non-saturating limits.

3.8. Multi-Contingency Sensitivity Sweep Across Inertia Regimes

To test beyond nominal grid parameters ($H = 3.20$ s, 5% contingency), Table 3 documents a 9-scenario factorial sweep. It includes small (3%, 9 MW), medium (5%, 15 MW), and severe (8%, 24 MW) generation trips across low ($H = 2.00$ s), nominal ($H = 3.20$ s), and high ($H = 5.00$ s) inertia systems.

The parameter sweep indicates two operational trends:

Table 3: Multi-contingency sensitivity sweep across grid inertia regimes (6.0 s evaluation).

Grid Inertia H (s)	Trip ΔP_L (%)	Base Nadir (Hz)	PULSE Nadir (Hz)	Nadir Lift (mHz)	Relative Lift (%)	PULSE Msgs / Agent
2.00	3% (9 MW)	59.5829	59.7451	+162.2	+38.9%	6.35
2.00	5% (15 MW)	59.3049	59.5172	+212.3	+30.5%	10.18
2.00	8% (24 MW)	58.8878	59.1037	+216.0	+19.4%	11.34
3.20	3% (9 MW)	59.6535	59.7695	+115.9	+33.4%	4.83
3.20	5% (15 MW)	59.4225	59.5918	+169.2	+29.3%	8.58
3.20	8% (24 MW)	59.0761	59.2525	+176.4	+19.1%	10.62
5.00	3% (9 MW)	59.7106	59.7951	+84.5	+29.2%	3.67
5.00	5% (15 MW)	59.5176	59.6460	+128.4	+26.6%	6.50
5.00	8% (24 MW)	59.2282	59.3746	+146.5	+18.9%	7.97

1. **Value in Low Inertia Settings:** Because PULSE contributes a fixed virtual inertia $H_v = 1.333$ s, its relative addition expands at lower system inertia: +66.7% at $H = 2.00$ s ($H_{\text{eff}} = 3.333$ s), compared with +41.7% at $H = 3.20$ s and +26.7% at $H = 5.00$ s. For a 5% trip, nadir improvement is +212.3 mHz at $H = 2.00$ s (59.3049 $\rightarrow$ 59.5172 Hz), +169.2 mHz at $H = 3.20$ s, and +128.4 mHz at $H = 5.00$ s.
2. **Autonomous Traffic Scaling:** For 8% trips (24 MW loss), the initial transient produces 10.62–11.34 messages per agent over 6 s. For 3% trips, message volume drops to 3.67–6.35. Event rates scale naturally with disturbance severity without supervisory threshold adjustments.

3.9. Heterogeneous Rack Level Battery Fleet Analysis

Rack level batteries exhibit varied initial state of charge and capacity due to staggered cell commissioning and thermal gradients across server rows. We evaluate a heterogeneous fleet of $N = 200$ agents where:

- Initial states of charge are distributed via $\text{SoC}_i(0) \sim \mathcal{U}(75.0\%, 95.0\%)$ (mean 84.98%).
- Electrochemical pack capacities are degraded via $E_{\text{nom},i} \sim \mathcal{U}(25.0, 30.0)$ kWh (mean 27.47 kWh, representing up to 16.7% capacity fade from nominal 30.0 kWh).

During a 30 s response to the 5% generation contingency, the heterogeneous fleet reaches a nadir of 59.5918 Hz, matching homogeneous fleet performance. The local gating function $a_i(t) = \text{clip}((\text{SoC}_i - 80.0\%)/10.0\%, 0, 1)$ shifts injection duty toward packs with higher available capacity, while packs near the 80.0% boundary taper output to safeguard server loads. Post-event autonomous IT runtime averages 46.76 minutes, with the most degraded pack retaining 40.00 minutes. All 200 agents satisfy Tier III (10.0 min) and Tier

IV (15.0 min) requirements with 100% compliance, while modeled 10-year capacity fade increment remains at 0.01795%.

4. Discussion and Practical Deployment Considerations

The simulations and stability derivations highlight practical trade-offs:

4.1. Decentralized Communication Scalability

SCADA polling at 100 ms generates constant message overhead, whereas PULSE transmits zero packets during normal steady state operation. During disturbances, adaptive threshold relaxation cuts active traffic by 63.6% relative to fixed threshold schemes. Because each agent responds solely to local measurements, communication links avoid central aggregation bottlenecks.

4.2. Physical Sensor Dynamics vs Idealized Formulations

Studies assuming unfiltered measurements report response times below 10 ms. In physical hardware, PLL and PMU filters require finite measurement windows (16.7–100 ms). Our unfiltered benchmark achieves 90% power injection in 7.3 ms, but this idealized speed omits measurement filtering. With 1-cycle PLL filtering, simulated rise time is 65.0 ms; under 2-cycle PMU filtering, rise time is 90.5 ms. Both filtered configurations arrest frequency decline without the derivative chatter observed in the unfiltered high-gain regime.

4.3. Stability Margins and Network Delays

Frequency domain derivations confirm an open loop phase margin of 68.2°. The idealized regime provides a delay margin of 1.58 s, while nominal filtered control retains a 0.952 s delay margin. Non-linear time domain simulations confirm closed loop stability under 30% packet drops and 50 ms latency jitter.

4.4. Data Center Mission Alignment and Sustainability Impact

Grid frequency support must not compromise IT server availability. In our battery model, a 30 s frequency event consumes 0.166 kWh per rack (33.1 kWh fleet total). Mean post-contingency backup runtime remains at 46.8 min, with a 40.0 min minimum floor, confirming 100% compliance with Tier III and Tier IV criteria. Projected 10-year capacity fade increment is 0.018% under the simulated duty profile. Re-purposing idle UPS storage for primary frequency containment directly supports UN Sustainable Development Goals 7 and 11.

5. Conclusions

Decentralized UPS fleet control arrests grid frequency decline following generation contingencies while preserving required backup energy. In the 300 MVA benchmark system, a 200-agent, 12 MW UPS fleet raises the frequency nadir from 59.423 Hz to 59.721 Hz following a 5% generation trip, matching continuous droop performance (59.731 Hz). The fleet supplies $H_v = 1.333$ s (+41.7%) virtual inertia and curbs 100 ms and 500 ms RoCoF values from -0.465 Hz/s and -0.441 Hz/s to -0.392 Hz/s and -0.285 Hz/s, respectively. PULSE emits zero messages during ambient grid drift and reduces contingency message counts by 63.6% relative to fixed threshold control. Incorporating 1-cycle PLL and 2-cycle PMU measurement filters yields closed loop 90% rise times of 65.0 ms and 90.5 ms, eliminating derivative chatter. Frequency domain stability derivations yield an analytical delay margin of 0.952 s under nominal filtering (1.58 s unfiltered), exceeding network latencies. Electrochemical modeling confirms that a 30 s support event uses 0.166 kWh per rack, maintaining over 40 min of autonomous backup runtime and achieving 100% compliance with Tier III and Tier IV reliability standards.

CRediT authorship contribution statement

Seyed Mahmoud Sajjadi Mohammadabadi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No human participants or real patient data were used. All code, random seeds, and scripts needed to reproduce the tables and figures will be made publicly available upon acceptance and are available to the editor and reviewers during review upon request. The simulation code, verification suites, and benchmark datasets are publicly available in the GitHub repository at `https://github.com/mahmoudsajjadi/PULSE.git` and archived persistently in Zenodo [24] at `https://doi.org/10.5281/zenodo.14920188`.

Declaration of Generative AI and AI-Assisted Technologies

AI assisted with language and grammar checking. The author reviewed all output and independently verified the computational results, taking full responsibility for the article.

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